Gnan Reddy Bobba

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About

Aspiring Software Engineer having keen interest in building scalable backend systems.

Technical Skills

Languages: Go, Python, SQL
Backend: chi, PostgreSQL, Apache Kafka
Infrastructure: Docker, MinIO (S3)
Tools: Git, GitHub, Visual Studio Code

Projects

link | **multimodal-rag-system** Present
Go, Python, PostgreSQL (pgvector), Apache Kafka, MinIO (S3), Docker

- Building a MVP of LLM based interface event-driven Multimodal RAG system which ingests text and images, answers questions with source citations.
- Currently it's under progress and is supervised by Dr. T. V. Gopal.

link | **webhook-delivery** Oct 2026
Go, PostgreSQL, chi, Docker

- Built a Postgres-only webhook delivery service in Go that fans out events, signs requests with HMAC-SHA256, and retries up to **8 times** with exponential backoff via `FOR UPDATE SKIP LOCKED`.
- Engineered per-endpoint isolation (concurrency caps, circuit breakers) so one failing endpoint can't stall others, sustaining **~3,230 deliveries/s** across **16 endpoints** — **2.7x** the ~1,202/s single-endpoint ceiling.
- Hardened against untrusted destinations with SSRF protection and AES-256-GCM-encrypted secrets at rest, verified by a `go test -race` suite covering crash recovery and idempotent duplicate-event handling.

link | **matscan** Sept 2026
Go

- Built a CLI-based matrix scanner for parsing and processing matrix input.

link | **calculator-cli** Apr 2026
Go

- Built a CLI-based calculator in Go supporting standard arithmetic operations.

Achievements

- Topped CBSE Class XII 2020 Mathematics at Delhi Public School, Nacharam with 95% marks
- Awarded a Rs. 4,000 cash prize by GAIL (India) Ltd. for academic distinction in Class XII 2020

Education

Mahindra University Hyderabad
M.Tech in Artificial Intelligence & Data Science *2025 - Present*

- **Relevant Coursework:** Machine Learning, Data Science, Artificial Intelligence, Computer Vision

Mahindra University Hyderabad
B.Tech in Computer Science & Engineering *2021 - 2025*

- **Relevant Coursework:** Data Structures and Algorithms, Operating Systems, Database Management Systems, Computer Networks